



HANDOFF

Agent module — next developer notes

Capabilities · contracts · codes that must work · execution checklist

Merge rule: everything lives in `/agent` only. Host wiring is a separate one-line change (see `INTEGRATION.md`). Do not edit files outside `/agent` for this merge. Product reference (static docs): codzure-solutions.vercel.app

1. Read in this order

1. `contract.ts` — shapes + ports (dependency-free). **Do not break.**
2. `INTEGRATION.md` — `registerAgent(mount, deps)` + `Stub` → `Real`.
3. `adapters/real.ts` — fill every `// TODO`.
4. `README.md` — run / curl / browser chat.
5. This document — capabilities + acceptance codes.

2. Capabilities (must remain)

Product	Capability	Tool	Behavior
Neo	Draft sale from NL	<code>record_sale</code>	Returns <code>requiresConfirmation: true</code> — no write
Neo	Confirm draft	<code>confirm_action</code>	Executes by <code>draftId</code> → persists <code>Sale</code>
Neo	List sales	<code>list_sales</code>	Read-only, immediate
Nyumba	Search listings	<code>search_listings</code>	<code>location</code> / <code>bedrooms</code> / <code>maxPrice</code> / <code>nearby</code> — immediate

Also: browser chat `GET /agent` (Codzure logo), `health/tools`, LLM providers `local` (default free) · `openai-compatible` · `anthropic`, `StubAdapters` + `RealAdapters` template.

3. HTTP that must keep working

Routes under `/agent/*` only. Env: `AGENT_*`.

Method	Path	Must return
GET	/agent	HTML chat
GET	/agent/assets/codzure_logo_full.png	PNG 200
GET	/agent/health	{ ok, module, products, llm, demoMode, chat }
GET	/agent/tools?product=neo nyumba	Tool list
POST	/agent/message	AgentResponse

POST /agent/message — request (must not change)

```
{
  "message": "string",
  "product": "neo" | "nyumba",
  "userId": "string",
  "sessionId": "string"
}
```

Response shape

```
{
  "reply": "string",
  "product": "neo" | "nyumba",
  "sessionId": "string",
  "draft": {
    "id": "agent_draft_...",
    "tool": "record_sale",
    "summary": "...",
    "requiresConfirmation": true
  },
  "data": {}
}
```

`draft` optional (writes only). `data` optional structured tool output.

4. Ports that must work (`contract.ts`)

RecordSalePort

- `draftSale(input) → { requiresConfirmation: true, draftId, summary }` — must **not** write yet

- `listSales(userId?)` → `Sale[]`

SearchListingsPort

- `search({ location, bedrooms?, maxPrice?, nearby? })` → `Listing[]` — immediate

ConfirmActionPort

- `getDraft(draftId)` → `DraftAction | null`
- `confirm(draftId)` → success/error — **this** writes Neo sales

Guardrails: draft-only writes · confirm by id · reads immediate · no singletons/monkey-patches · agent_ id prefix.

5. Execution codes that must pass

With StubAdapters + `AGENT_LLM_PROVIDER=local` :

Neo

```
# 1) Draft — must return draft.requiresConfirmation === true
curl -s -X POST http://127.0.0.1:8787/agent/message \
  -H 'Content-Type: application/json' \
  -d '{"message":"sold 3 bags cement to Mary
1500","product":"neo","userId":"demo-user","sessionId":"neo-1"}'

# 2) Confirm — replace DRAFT_ID
curl -s -X POST http://127.0.0.1:8787/agent/message \
  -H 'Content-Type: application/json' \
  -d '{"message":"confirm DRAFT_ID","product":"neo","userId":"demo-
user","sessionId":"neo-1"}'

# 3) List
curl -s -X POST http://127.0.0.1:8787/agent/message \
  -H 'Content-Type: application/json' \
  -d '{"message":"show my sales","product":"neo","userId":"demo-
user","sessionId":"neo-1"}'
```

Nyumba (search must return listings)

```
curl -s -X POST http://127.0.0.1:8787/agent/message \
  -H 'Content-Type: application/json' \
  -d '{"message":"find 2 bedroom places in Westlands under 100000 near
Sarit","product":"nyumba","userId":"demo-user","sessionId":"ny-1"}'
```

```
curl -s -X POST http://127.0.0.1:8787/agent/message \
  -H 'Content-Type: application/json' \
  -d '{"message":"show land in Ruiru","product":"nyumba","userId":"demo-user","sessionId":"ny-2"}'
```

Smoke + compile

```
curl -s http://127.0.0.1:8787/agent/health
curl -s 'http://127.0.0.1:8787/agent/tools?product=neo'
cd agent && npm install && npm run typecheck && npm run build
```

6. LLM providers

AGENT_LLM_PROVIDER	Paid?	Notes
local (default)	No	Weak local NLU → same tools
openai-compatible	Free tier OK	Groq etc. via BASE_URL + API_KEY + MODEL
anthropic	Credits	Needs AGENT_ANTHROPIC_API_KEY with balance

7. Host wiring (minimal)

```
import { registerAgent, createStubAdapters } from "./agent/index.js";

registerAgent((path, method, handler) => {
  app[method.toLowerCase()](path, async (req, res) => {
    const out = await handler({ body: req.body, query: req.query, headers: req.headers });
    res.status(out.status).type(out.contentType ?? "json").send(out.body);
  });
}, {
  ...createStubAdapters(), // → createRealAdapters() in prod
  anthropicApiKey: process.env.AGENT_ANTHROPIC_API_KEY ?? "unused",
  llmProvider: process.env.AGENT_LLM_PROVIDER ?? "local",
  demoMode: process.env.AGENT_DEMO_MODE === "1",
});
```

8. Pre-merge checklist

- ☐ npm run typecheck + npm run build green
- ☐ Neo draft → confirm → list works

- ☐ Nyumba Westlands + Ruiru search returns listings
- ☐ GET /agent chat loads with Codzure logo
- ☐ adapters/real.ts TODOs planned / done for prod
- ☐ Env only AGENT_* ; routes only /agent/*
- ☐ Write tools still require confirmation
- ☐ No edits outside /agent
- ☐ Secrets never committed

9. File map

```
agent/  
  contract.ts          ← MUST stay stable  
  knowledge.ts  
  index.ts             ← registerAgent / handleAgentMessage  
  demo-ui.ts / assets/  
  core/loop.ts / weak-llm.ts / openai-compat.ts / registry.ts / logger.ts  
  adapters/stub.ts / real.ts  
  tools/neo/* / tools/nyumba/*  
  HANDOFF.md · HANDOFF.html · HANDOFF.pdf  
  INTEGRATION.md · README.md
```

Codzure Solutions · Neo (NeoBuk) & Nyumba Zetu (Ask Nyumbani) · Agent module handoff

Source: agent/HANDOFF.md · PR branch `cursor/agentic-ai-module-98e3`